



Certified Pre-Owned BBA SUV Finder

—Comparing BMW X3, Audi Q5, and Mercedes-Benz
GLC 300 using real listing data

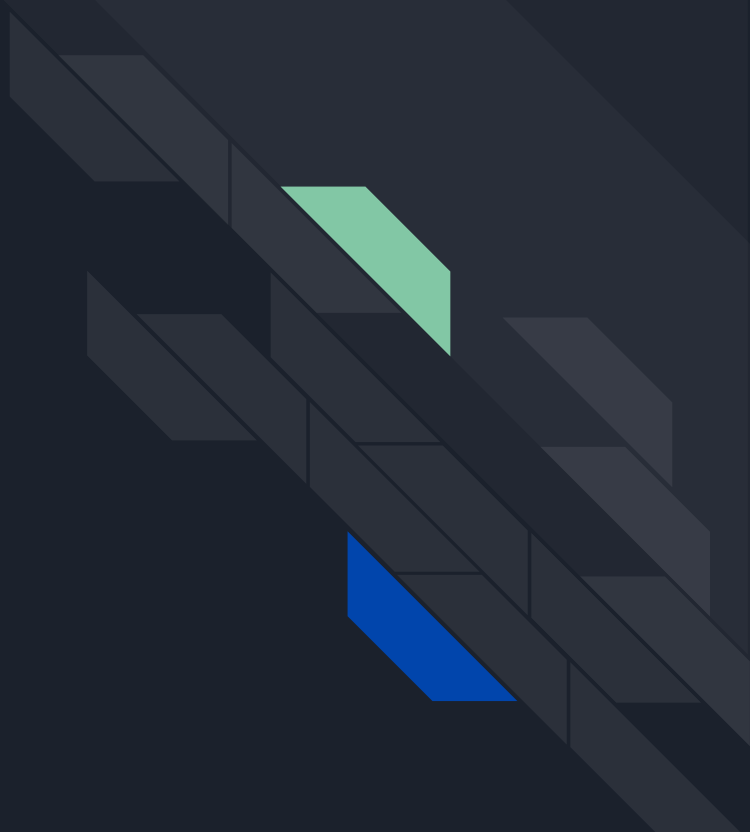
Yufei Meng

My Problems

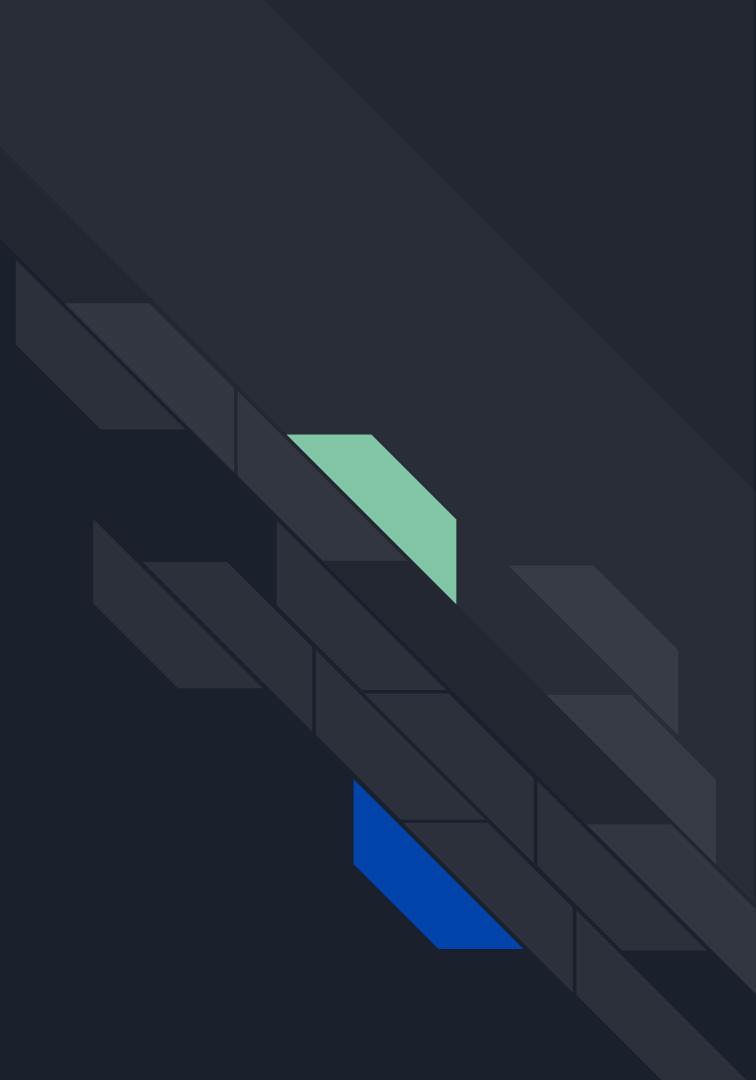
- How much should a CPO BBA SUV cost?
- Can the basic vehicle characteristics help predict listing price?
- If so... how?

And...Motivation

- I wanted to see if I could use some real listing data to understand the price. And of course, to find some cars that match my preferences.
- Without spending tons of time on browsing dealers' websites...



Data Collection & Cleaning



Source: Auto.Dev API

- BMW X3: 95
- Benz GLC 300: 24
- Audi Q5: 29

- BMW X3: 95
- Benz GLC 300: 24 → 21
- Audi Q5: 29

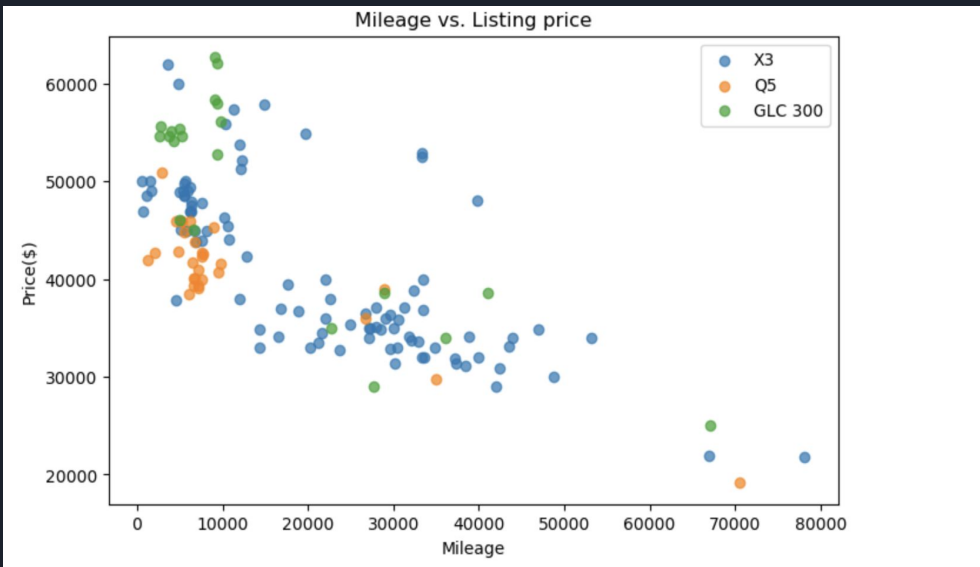


- CPO only
- San Diego - Zip 92122
- 200-Mile Radius

INSANE Price🤔:

\$85 for a Benz GLC 300

Exploratory Data Analysis



- GLC 300s listings were generally listed at higher prices.
- Higher mileage was associated with lower listing prices.
- Newer model years tended to have higher prices.

But these factors overlap — so average prices alone don't tell the whole story.

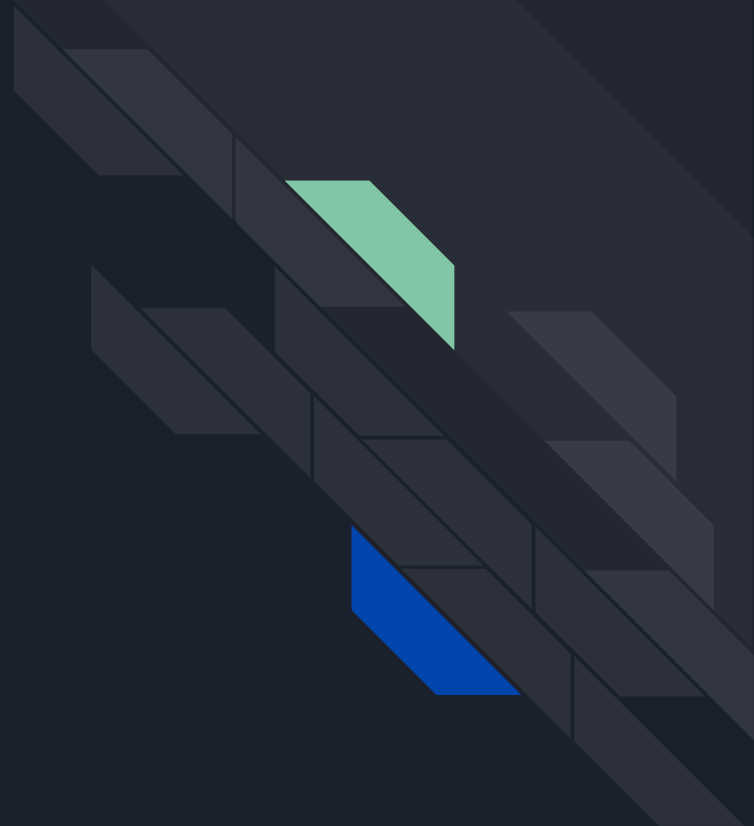
Regression Analysis

Prediction:

Price ~ Mileage + Model Year + Vehicle Model

Result:

- **-\$2,113** per +10,000 miles
- **+\$3,208** per +1 model year
- **$R^2 = .811$** on test data
- **MAE $\approx$ \$2,881**





CPO SUV Finder: From Analysis to a Usable Tool

```
Welcome. What's the brand you are looking for? benz
What's your maximum budget? 50000
What's your expectation for the maximum mileage? 30000
What's the expectation for the minimum model year? 2023
How would you like to sort the results? (price / mileage): price
```

Found 4 matching cars:

	year	make	model	trim	price	miles	dealer	city	vin	listing_url
144	2023	Mercedes-Benz	GLC 300	4MATIC	34998	22721	crown lexus	Ontario	W1NKM4HB1PF026908	https://crownlexus.com/certified/Mercedes-Benz...
141	2023	Mercedes-Benz	GLC 300	GLC 300	38596	28928	crown lexus	Ontario	W1N0J8EB8PG165682	https://crownlexus.com/certified/Mercedes-Benz...
143	2026	Mercedes-Benz	GLC 300	GLC 300	44999	6628	crown lexus	Ontario	W1NKM4GBXTF581853	https://crownlexus.com/certified/Mercedes-Benz...
142	2026	Mercedes-Benz	GLC 300	GLC 300	45999	4997	crown lexus	Ontario	W1NKM4GB8TF578868	https://crownlexus.com/certified/Mercedes-Benz...

This Finder can filter by:

- Brand
- Maximum budget
- Maximum mileage expectation
- Minimum model year expectation
- And! Sort by price or mileage

Instead of browsing listings manually,
the finder could turn the collected data
into personalized results.

Thank you! ❤️

